# Wetland_emissions_r2.R

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`SOCCR_Wetlands` *Create SOCCR based wetland methane maps*

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### Description

'SOCCR_Wetlands' writes up to 3 netcdf files of gridded wetland methane emissions, 1 for SOCCR1, 1 for SOCCR2, and 1 for freshwater. Includes optional visuals as well.

### Usage

```
SOCCR_Wetlands(
  output_directory,
  plot_directory,
  domain,
  domain_template,
  Use_SOCCR1,
  Use_SOCCR2,
  Include_freshwater,
  Wetland_EFs,
  verbose,
  County_Tigerlines,
  State_Tigerlines,
  watershed_shapefile
)
```

### Arguments

`output_directory`
: Character providing the full filepath to save processed data

`plot_directory`
: Character providing the full filepath to save figures. Only relevant if verbose = TRUE.

`domain`
: SpatVector polygon outlining the desired output area

domain_template
: SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system

Use_SOCCR1
: Logical. Pulled from config file. Indicating whether or not to calculate emissions using SOCCR1.

Use_SOCCR2
: Logical. Pulled from config file. Indicating whether or not to calculate emissions using SOCCR2.

Include_freshwater
: Logical. Pulled from config file. Indicating whether or not to calculate emissions for freshwater wetlands using Rosentreter et al.

Wetland_EFs
: Data frame. Pulled from config file. Emission factors to use for all wetlands - including SOCCR1, SOCCR2, and Rosentreter et al.

verbose
: Logical indicating whether to save additional output. This includes plots of the gridded methane emissions on log scales, saved separately for SOCCR1, SOCCR2, and freshwater.

County_Tigerlines
: SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html. Only relevant if verbose=TRUE.

State_Tigerlines
: SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html. Only relevant if verbose=TRUE.

watershed_shapefile
: Character. Commission for Environmental Cooperation watershed shapefile. Available at http://www.cec.org/north-american-environmental-atlas/watersheds/. Only relevant if USE_SOCCR2 = TRUE.

## Details

This function takes the output of ‘NWI_Wetland_fraction‘, which is per pixel fractional coverage of wetlands separated by wetland type, and applies emission factors to convert coverage to methane emissions. It is simply applying SOCCR1 or SOCCR2 average emissions from wetlands to the NWI activity data. Additionally, freshwater emissions can be calculated using emissions from Rosentreter et al.

SOCCR1 values are based on the arithmetic averages of Table F5, SOCC2 values are based on the arithmetic averages of Tables 13B.8 to 13B.11 for PFO and PNF and Table 15A.2 for M2 and E2, and Lakes and Rivers (L1, L2, and R1 - R4) are from Rosentreter et al. using the median flux from rivers and the largest lake class (>1 km). SOCCR2 is calculated by watershed as there was regionally separated data. The lake flux was chosen as McDonald et al. show that large lakes (>1 km2) constitute 71% of the total lake area in CONUS (rising to 90% if including the Great Lakes).

SOCCR1 is available at https://www.carboncyclescience.us/state-carbon-cycle-report-soccr and SOCCR2 is available at https://carbon2018.globalchange.gov/.

See references McDonald et al. and Rosentreter et al.

## Value

Nothing is returned from the function, but the main outputs are up to 3 netcdcf files of the methane emissions from wetlands with 1 file for SOCCR1 and 1 file for SOCCR2 based emissions. Lakes and rivers from Rosentreter et al. emissions can also be included. They are titled "SOCCR1.nc", "SOCCR2.nc", and "Freshwater.nc".

If verbose is set to TRUE, then multiple figures are also saved. Log scale plots with consistent axes are saved for the 2 SOCCR emissions and freshwater emissions. They are saved as "SOCCR1.png", "SOCCR2.png", and "Freshwater.png".

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## Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
             ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
             xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
             crs=grid_crs)
grid_vect <- as.polygons(ext(grid),crs=grid_crs)
EFs <- data.frame("E2"=c(10.3,15.29*16.043/12.011),
                          "M2"=c(10.3,15.29*16.043/12.011),
                          "PFO"=c(36,18.52*16.043/12.011),
                          "PNF"=c(36,24.92*16.043/12.011),
                          "L1"=5,
                          "L2"=5,
                          "R1"=7.88,
                          "R2"=7.88,
                          "R3"=7.88,
                          "R4"=7.88)
rownames(EFs) <- c("SOCCR1","SOCCR2")

# convert from g CH4 per m2 per yr to nmol/m2/s
EFs=EFs*1E9/(16.043*365.25*24*60*60)

SOCCR_Wetlands(output_directory="~/../Desktop/out/",
               plot_directory="~/../Desktop/plots/",
               domain=grid,
               Use_SOCCR1=TRUE,
               Use_SOCCR2=TRUE,
               Include_freshwater=TRUE,
               Wetland_EFs=EFs,
               verbose=TRUE,
          County_Tigerlines=vect("~/../Desktop/in/County_Tigerlines/tl_2018_us_county.shp"),
          State_Tigerlines=vect("~/../Desktop/in/State_Tigerlines/tl_2018_us_state.shp"),
          watershed_shapefile="~/../Desktop/in/watersheds_shapefile/watershed_p_v2.shp")
```

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